

# Exploratory and statistical data analysis of Indian economy

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**Abstract**— Exploratory Data Analysis provides visual exploration of data and also allows for the identification of trends and relationships between variables. The aim of this research is to study about different factors involved in the economy of India. The dataset has been collected from year 1960 to 2020 and is analyzed by looking at the factors such as GDP, import and export value of goods and services, total reserves(include gold, current US Dollar) and health status of nation's citizens. Thus, it suggests that all factors are important for a stable economy and it also provides different ways to improve these factors.

**Keywords**—Exploratory Data Analysis (EDA), Indian Economy, Correlation

## I. INTRODUCTION

Exploratory Data Analysis (EDA) is an approach that involves summarizing and representing data effectively by emphasizing its essential characteristics. EDA primarily concentrates on tasks such as handling missing data, adjusting variables when required, and verifying the assumptions essential for constructing models and testing hypotheses. In the process of EDA, various visualizations like data distributions, bar charts, box plots, and histograms are employed to identify potential correlations or relationships between variables [1].

The gross domestic product (GDP) is the comprehensive measure of a nation's economic well-being, representing the total value of all goods and services produced within a country's borders during a specific time frame. It serves as a broad gauge of a nation's overall economic performance and reflects its entire domestic output [2]. Gross domestic product, or GDP, is a measure of the size and health of our economy as a whole. The entire market worth of all the goods and

services produced in a year is measured by the GDP. It is frequently regarded as the best indicator of a country's effective performance.

Exports can be defined as the items which are sent to other countries to get buyers in another. Export agreements are often highly strategic, involving countries entering into arrangements to enable their own nation to export goods for increased domestic revenue, while also securing access to the products they require through imports [3].

An import can be defined as the items which are bought from other country. Countries import those items which cannot be produced in their country or the items which are not available in their country [4].

GDP is the sum up of all these economic activity in a certain period. The objectives of this paper are:

1. To find the important factors involved in the GDP
2. To find their role in the GDP
3. To find the ways to improve these factors

The organization of paper is as follows: First section 'Introduction' provides insights on Indian economy. In 'Literature Survey' a brief review towards some related work related to data analysis of Indian economy is discussed. Different pre-processing techniques and data analysis used in this study are presented in section 'Methodology'. Detailed discussion about data analysis and visualization is presented in section 'Data analysis and visualization'. Section 'Conclusion' concludes the paper.

## II. LITERATURE SURVEY

Before 1991, India was struggling with its economic crisis adopts the policy of Liberalization, Privatization and globalization to perform better. The first few plans were to focus on manufacturing and industrial sector to have a strong backbone [5].

FDI growth negatively correlated with Trade balance, export less than import and to achieve positive correlation between Trade balance and FDI, export more goods and reduce import [6].

In [7] authors have provided a model that has been able to track the historical movements of the variable accurately. The Trade, Transport, Storage and Communication sector is found to have the greatest impact on the GDP trend under study, while the Financing, Insurance, Real Estate and Business Services sector has the least impact.

India performance increased over time because of rapid expansion of service producing industries. Another factor is modest level of investments. Manufacturing sectors are not solely important [8].

In [9] authors have investigated the relationship between COVID-19 and its connection to economic growth, as assessed through gross domestic product (GDP). Furthermore, the research explores the association between COVID-19 and the top five alternative sources of tax revenue, which include the S&P 500 (GPSC), crude oil, gold, natural gas, and the 20+ Year Treasury Bond.

The research also demonstrates the relation between crude oil and whole sale price index and how their prices affect Indian economy [10].

## III. METHODOLOGY

This section discusses various methods used for the analysis purposes. The flowchart of EDA is shown in fig. 1. The steps are:

### A. Dataset collection

In this paper, dataset of GDP taken in csv format that has been downloaded from Kaggle.

### B. Data pre-processing

First of all, we perform data cleaning for that we check that dataset do not contain any null value so that it will not create problem in later steps.

### C. Exploratory data analysis

For data visualization a regression plot is plotted between import value & GDP, export value & GDP and import & export to find relationships/dependencies among them. A set of independent variables are used to predict a continuous (dependent) variable using the parametric technique of regression.

For statistical analysis heat map is plotted to find correlation among all the parameters involved. A Heat map is a 2-D data representation form of a map in which data values are represented as colors. It is use to show relationships between variables.

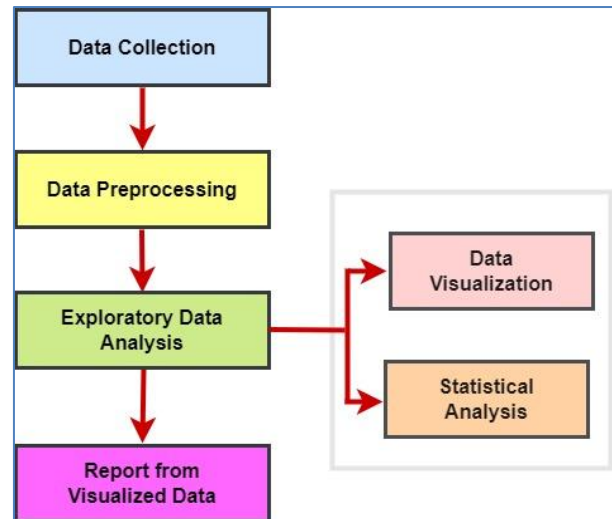


Fig. 1. Flowchart of Exploratory Data Analysis

## D. Result

Results are provided to find out the factors that are essential to make our GDP stronger and to reach pinnacle of success to become world largest economy.

## IV. DATA ANALYSIS AND VISUALIZATION

A Heat map is a 2-D data representation form of a map in which data values are represented as colors. It is used to show relationships between variables. Here we can find the correlation among the variables.

Correlation is a statistical method used to access a possible linear association between two or more variables.

From fig.2, i.e. heat map we can infer the following points to analyze the factors using heat map mentioned below:

We know that import and export plays a crucial role to find out the progress of a country.

Correlation of import and annual GDP = 0.3. This shows that the import of our country is more which makes our progress to get slowdown. It is essential for a country to maintain their reserves so that they can deal with unforeseen circumstances.

Correlation of GDP and Total reserves = 0.05. Thus, we can deduce that we need more foreign reserve to combat with miserable situations. It will also help to deal with unexpected global crises so that our country will not be missed out from any necessary need.

Population is an asset for a nation. Thus, we need a healthy and fit population. Correlation of life-expectancy-at birth & population growth = -0.9. Thus, we need to work on the health of the citizen so that they can be an asset for country not a liability. We need to work for providing medications, supplements and healthy food in order to increase their life expectancy at birth. This will combat the issue of liable population and will increase the sustainedly in life.

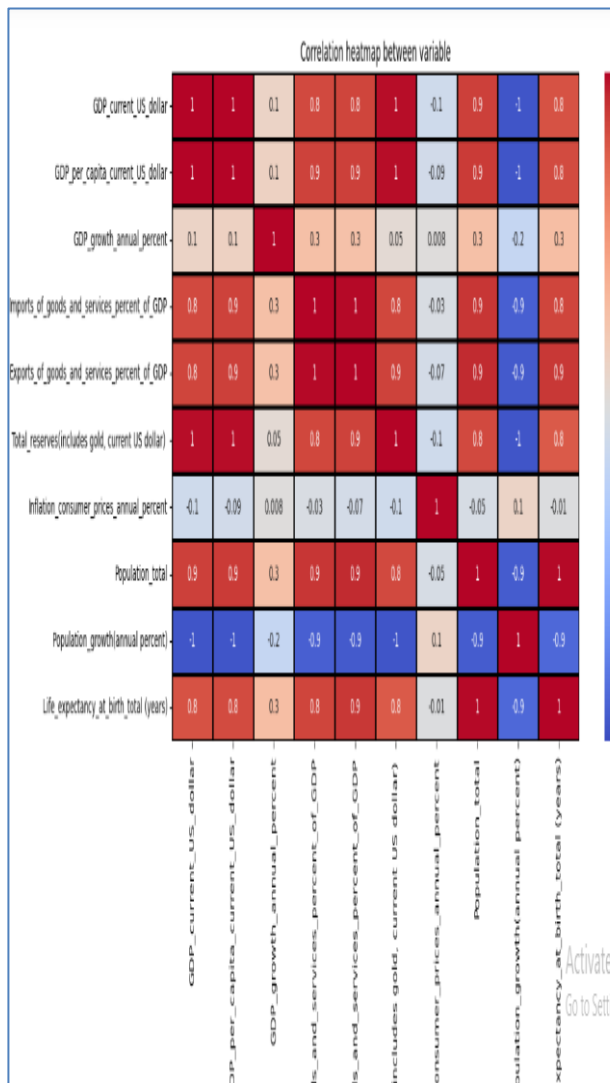


Fig. 2. Heat map

From fig.2, we can infer the plot between the export and GDP to analyze the relation between them. The aim is to analyze the functional dependencies of both the factors of the mentioned above.

A regression plot is drawn between GDP and export in order to statistically analyze the factors and to find the governing factor in between them. A set of independent variables are used to predict a continuous (dependent) variable using the parametric technique of regression.

From fig.3, we can infer the plot between the import and GDP to analyze the relation between them. The aim is to analyze the functional dependencies of both the factors of the mentioned above.

From fig.4, we can study the relation between the import and export of the country. We can find the functional dependencies between them

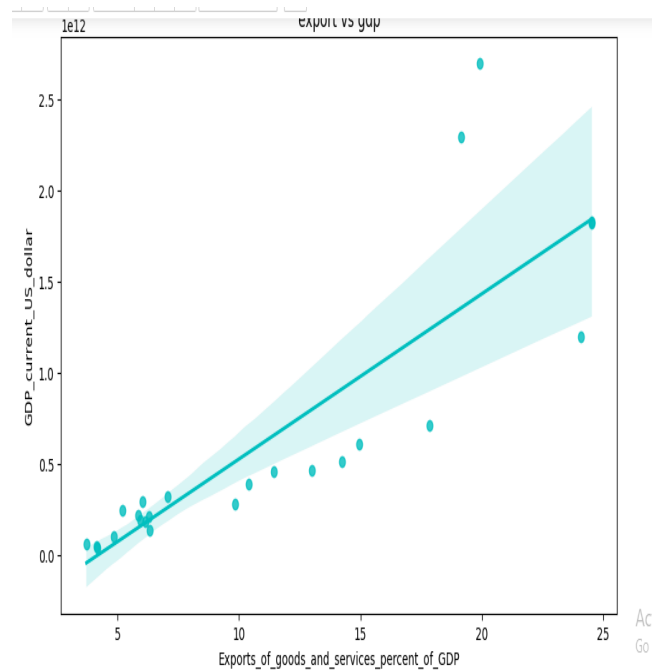


Fig. 3. GDP VS Export regression plot

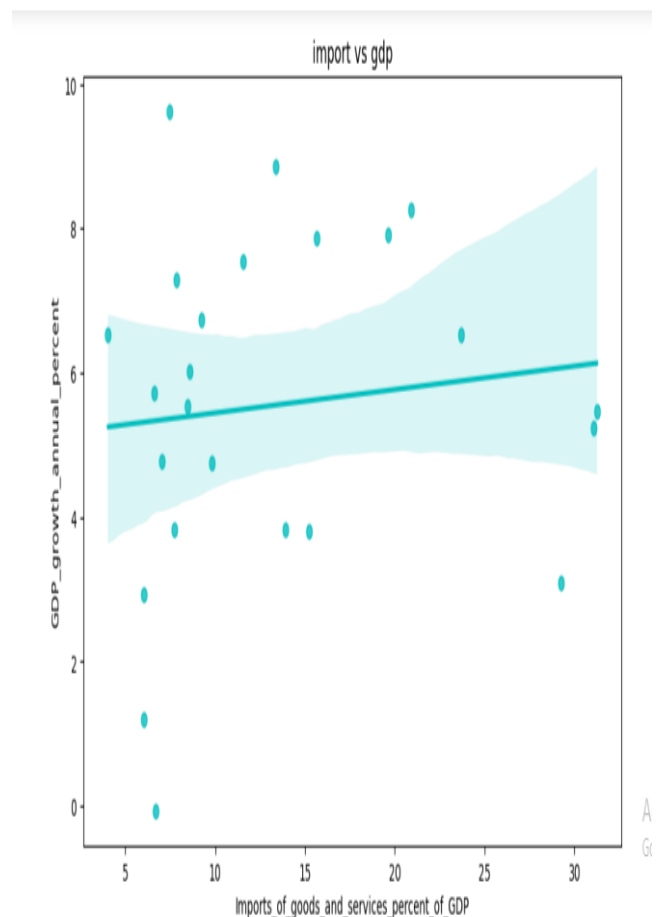


Fig. 4. GDP VS Import regression plot

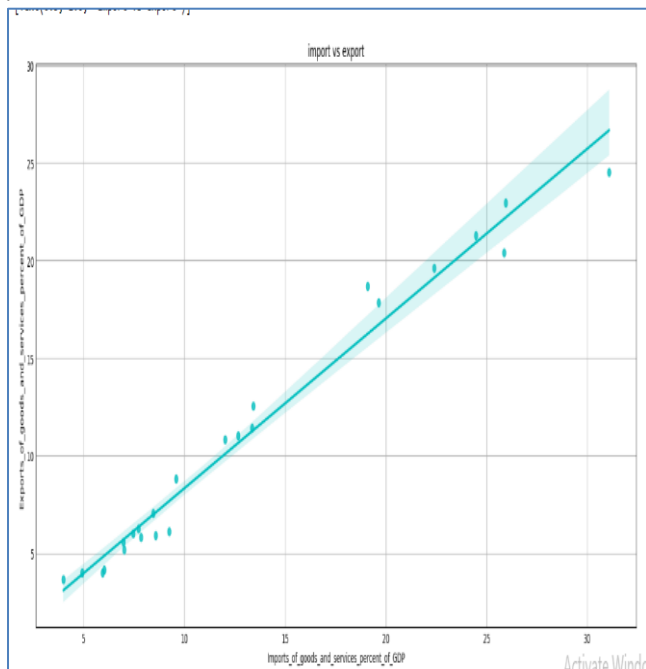


Fig. 5. Export VS Import regression plot

From fig.2, and fig.3, we can infer that if import of a country is greater than the export of a country. This will be major concern. Since, it will affect the Indian Economy.

## V. CONCLUSION

Analysis of Indian Economy has been performed and it has been analyzed that when import of a country is greater than export, this means, they have negative Balance of Trade (BOT). In order to have a positive Balance of Trade the export of a country must be greater than the import of the country.

As per the study, the import of petroleum & its products, gold & other resources is very high. Since we have to rely on European and Arabian oil for a major proportion of our need. They have large reserves of oil. For other minerals and gold, we are facing the problem of less production and quality of minerals which are not sufficient to meet our demand. There is a need to establish more manufacturing units to combat with such issues. Like, we can derive fuel from plastic waste in large amount if we have high Manufacturing units. India being an agrarian country lacks in production. Around 75% people are indulging in Agriculture. Small production units are present but not at large scale. With the help of techniques applied (linear regression) and dataset used the system does is best to identify the governing factors of Indian Economy.

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